

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

TherAlign Health, Founding Software Engineer Intern | *Remote* Jul 2026 – Present

- Built provider-facing **SMART on FHIR** decision-support workflows using **Next.js** and Firebase/**GCP**, surfacing guideline-backed medication alternatives during prescribing to reduce prior-authorization friction.
- Extended a **Cloud Functions** pipeline combining **RxNorm** normalization, formulary rules, and PubMed evidence; restricted **Gemini** to evidence synthesis while eligibility and coverage remained deterministic.

Culture & Morality Lab, UMass Amherst, Lab Assistant | *Amherst, MA* Jul 2026 – Present

- Sole engineer building the lab's CCR text-analysis platform (Python, **FastAPI**, sentence-transformers, React) under full technical ownership, productionizing a published NLP method into a self-serve web tool for researchers.
- Built a YAML model registry, a versioned construct library (**99 constructs from 38 questionnaires**), and a structured data-quality warning system; grew the backend suite to **40 tests** (hermetic CI).

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual annotation by **100+ hours/month** and lifting labeling throughput by **60%**.
- Owned backend security and reliability for the eTriage stack, implementing **AWS KMS** encryption for clinical PII, hardening authentication and rate limiting, and remediating **20+ VAPT findings** with regression-tested fixes.
- Served as primary technical point of contact for the eTriage backend; ran structured knowledge transfer that onboarded a second engineer to on-call readiness in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot productivity experiment across migrations and tech-debt PRs.

Projects

align-ops – LLM Post-Training + Serving Ops PyTorch, TRL (SFT/DPO/GRPO), vLLM, FastAPI, Docker

- Post-trained **Qwen2.5-1.5B** through SFT, DPO, and GRPO (QLoRA, **1.4%** of params), lifting held-out task accuracy **54%→72%** and reaching a **71% win-rate** over base across **500 eval prompts**; served it via **FastAPI + vLLM** with async micro-batching and Redis caching for **1.9× throughput**, **640 ms p95 TTFT**, and GPU utilization **52%→84%**.

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized C++ matrix multiplication with cache-aware tiling, x86 **AVX/SIMD** vectorization, register-aware micro-kernels, and **OpenMP** multithreading, profiling L1/L3/TLB behavior with **perf**.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Optimized a large-scale distributed Spark ETL pipeline over **~1.4B rows**, cutting end-to-end runtime **~25%** and the tail-latency ratio **2.14x to 1.74x** via broadcast joins, AQE, skew handling, and partitioning.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, C++, Java, TypeScript, JavaScript, SQL, Bash, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: FastAPI, Node.js, React, Next.js, Docker, Git, GitHub Actions, REST APIs

Systems / Distributed: data structures & algorithms, AVX/SIMD (x86), OpenMP multithreading, Spark / PySpark, distributed data processing, perf profiling, Linux

ML & AI: PyTorch, Hugging Face Transformers, TRL (SFT/DPO/GRPO), LoRA/QLoRA, vLLM, RAG, evaluation

Cloud / Platforms: GCP, AWS, Azure, Docker, Kubernetes, Prometheus/Grafana, MongoDB Atlas

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.